

Process Monitoring Dashboard

Real-time anomaly detection using LSTM Autoencoder + Ollama LLM

Refresh

Showing anomalies from latest entry minus 10 hours: Dec 31, 2024 09:57 PM to Jan 01, 2025 07:57 AM

Anomalies

3

High

0

Medium

0

Low

3

Avg Error

0...

Latest Anomalies Detected by Autoencoder

TIME	TOP CONTRIBUTING TAGS	RECONSTRUCTION ERROR
Jan 01, 2025 07:57 AM	- Composition_of_F_product_rate_of_change - Purge_rate_rate_of_change - Composition_of_B_purge_rate_of_change - Composition_of_C_purge_rate_of_change - A_feed_stream_rate_of_change	0.8861
Jan 01, 2025 04:27 AM	- E_feed_stream_rate_of_change - Separator_pot_liquid_flow_valve_rolling_mean - Separator_level_rolling_mean - Composition_of_B_rxtr_feed_rolling_mean - Composition_of_E_rxtr_feed_rate_of_change	0.9052
Jan 01, 2025 04:12 AM	- Separator_pot_liquid_flow_valve_rolling_mean - Separator_level_rolling_mean - Composition_of_B_rxtr_feed_rolling_mean - E_feed_stream_rate_of_change - Composition_of_F_product_rate_of_change	0.8914

LLM Root Cause Insights

Jan 01, 2025 07:57 AM

Low

Classification: uncertain

Affected Area: multiple areas

Summary: Multiple variables show deviations, but they are scattered across different units and do not show a clear pattern that would suggest a specific process area or failure mode. The reconstruction error is also close to the threshold, making it difficult...

Top Variables:

- Composition_of_F_product_rate_of_change (xmeas_39_diff)
- Purge_rate_rate_of_change (xmeas_10_diff)
- Composition_of_B_purge_rate_of_change (xmeas_30_diff)
- Composition_of_C_purge_rate_of_change (xmeas_31_diff)

Recommended Action: Investigate further by checking trends over time, correlation between variables, and consulting process knowledge for any known issues that might cause these variations.

Confidence: 0.3

Jan 01, 2025 04:27 AM

Low

Classification: uncertain

Affected Area: multiple areas

Summary: Variables outside p5-p95 are present, but they do not belong to the same process area or show a physically related pattern. Therefore, it is uncertain whether this is a likely real process anomaly.

Top Variables:

- E_feed_stream_rate_of_change (xmeas_3_diff)
- Separator_pot_liquid_flow_valve_rolling_mean (xmv_7_rollmean)
- Separator_level_rolling_mean (xmeas_12_rollmean)
- Composition_of_B_rxtr_feed_rolling_mean (xmeas_24_rollmean)

Recommended Action: Investigate further by analyzing more variables and their trends over time, considering inter-unit relationships and physical context.

Confidence: 0.5

Jan 01, 2025 04:12 AM

Low

Classification: uncertain

Affected Area: multiple areas

Summary: The Separator_pot_liquid_flow_valve_rolling_mean is outside p1-p99, but it's not related to any other variables in the same process area or showing a physically related pattern. The E_feed_stream_rate_of_change is also extreme high, but the Separator...

Top Variables:

- Separator_pot_liquid_flow_valve_rolling_mean (xmv_7_rollmean)
- E_feed_stream_rate_of_change (xmeas_3_diff)

Recommended Action: Investigate the E feed behavior and its impact on the separator and recycling loop. Also, check the cooling system for any changes that could affect the separator's operation.

Confidence: 0.4

